Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – PERIODIC TEST 08
Chapter 1: Real Numbers

Time: 60 Minutes
MM: 30

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based question of 4 marks with internal subparts.
7. Use of calculators is strictly prohibited. All mathematical steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. If two positive integers a and b are such that $a = 12$ and $\text{HCF}(a, b) = 4$, $\text{LCM}(a, b) = 24$, then b is equal to: [1]
(a) 8 (b) 6 (c) 16 (d) 10

Q2. The largest number which divides 70 and 125, leaving remainders 5 and 8 respectively, is: [1]
(a) 65 (b) 13 (c) 17 (d) 5

Q3. The largest number which divides 318 and 739, leaving remainders 3 and 4 respectively, is: [1]
(a) 105 (b) 15 (c) 35 (d) 210

Q4. The smallest number which when divided by 28 and 32 leaves remainders 8 and 12 respectively is: [1]
(a) 204 (b) 224 (c) 244 (d) 184

Q5. Three bells toll together at intervals of 9, 12, and 15 minutes respectively. If they toll together now, after how many hours will they toll together next? [1]
(a) 2 hours (b) 3 hours (c) 5 hours (d) 6 hours

Q6. **Assertion (A):** The HCF of two consecutive natural numbers is 1.
Reason (R): Two consecutive natural numbers are always co-prime. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. Two numbers are in the ratio 21 : 17. If their HCF is 5, find the numbers. [2]

Q8. Find the largest number that divides 2053 and 367, leaving remainders 5 and 7 respectively. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Find the HCF and LCM of 144, 180, and 192 by prime factorisation, and show that HCF×LCM ≠ Product of the three numb [3]

Q10. If $d = \text{HCF}(48, 72)$, find integers x and y such that $d = 48x + 72y$. [3]

SECTION D: Long Answer Type (2 × 5 = 10 Marks)

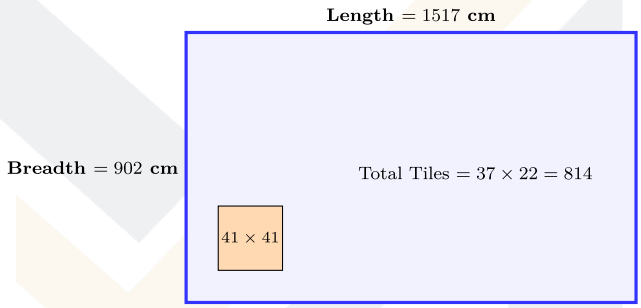
Q11. Prove that $\sqrt{5}$ is an irrational number. Hence, prove that $(3\sqrt{5} - 2)^2$ is an irrational number. [5]

Q12. Prove that if p is a prime number, then $\sqrt{p}$ is irrational. Hence, prove that $\sqrt{2} + \sqrt{5}$ is an irrational number. [5]

SECTION E: Case Study Based Question (1 × 4 = 4 Marks)

Q13. CASE STUDY: CASE STUDY 08: HERITAGE PALACE ARCHITECTURAL CERAMIC TILE PAVING [4 Marks]

An architectural restoration team is renovating a magnificent ceremonial banquet courtyard in a heritage palace. The rectangular courtyard measures 15 m 17 cm in length and 9 m 2 cm in breadth. The chief architect specifies that the entire courtyard must be paved with the minimum possible number of identical square ceramic tiles, cut without any material wastage.



(i) Determine the largest side length of the square tile in centimetres. [2]

(ii) Find the minimum number of tiles required to pave the entire courtyard. [1]

(iii) Calculate the perimeter of one single square tile in centimetres. [1]

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